

AKASH BAHADUR SINGH

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Summary

Backend Engineer with ~2 years of experience building scalable **Node.js/TypeScript** backend systems at enterprise scale. Shipped **SAML 2.0 SSO**, **Redis-cached API layers**, cron-based **REST integrations**, and data processing pipelines serving thousands of daily users — consistently driving measurable reductions in latency, incidents, and support overhead. Hands-on with **NestJS**, **MySQL**, **MongoDB**, and multi-tenant architecture design.

Experience

Truein

Aug 2024 – Present

Software Engineer (Backend)

Pune, Maharashtra

- Engineered **cron**-based REST integrations with Zoho Books, Hikvision, Entra ID, and Google Maps — 4,000+ daily enterprise users.
- Reduced API latency by **65%** via Redis caching and MySQL index optimization across high-traffic endpoints.
- Shipped **SAML 2.0 SSO** enabling federated auth for 10,000+ users across enterprise identity providers.
- Re-architected **idempotent** data pipelines processing 10,000+ daily records, cutting production incidents by 40%.
- Developed** automation workflows targeting recurring issue categories, reducing support tickets by **70%**.

Samsung Research and Development Institute

May 2023 – Jul 2023

Software Engineer Intern

Bangalore, Karnataka

- Analysed **4G/5G** telecom log data and authored Python scripts to extract network performance insights for fault detection across distributed systems.

Projects

Multi-Tenant SaaS Billing Engine | NestJS, TypeScript, Prisma, PostgreSQL, Redis, JWT, PDFKit | GitHub

- Architected **multi-tenant** billing engine with isolated tenant data models and subscription lifecycle — upgrades, downgrades, cancellations.
- Implemented idempotent** invoice generation and usage-based metering APIs preventing duplicate billing across tenants.
- Designed **Redis-backed async job queues** (BullMQ) decoupling background billing workflows from API request cycles.
- Enforced tenant-level data isolation via **NestJS interceptors and guards** with service/repository separation pattern.

Connect | Node.js, Express.js, React.js, MySQL, JWT | GitHub

- Created REST APIs with **JWT authentication**, relational schema design, and MySQL query optimization via indexing.

Skills

Languages: JavaScript, TypeScript, Python, C++

Backend: Node.js, NestJS, Express.js, REST APIs, BullMQ

Databases & Caching: MySQL, MongoDB, Redis, Prisma ORM

Auth & Security: SAML 2.0 SSO, JWT

Tools & DevOps: Git, GitHub, Jenkins, Linux, Swagger, Docker

Concepts: Data Structures & Algorithms, OOP, Computer Networks, Agile

Education

National Institute of Technology, Durgapur

Dec 2020 – May 2024

B.Tech in Computer Science & Engineering **CGPA: 8.44 / 10**

Durgapur, West Bengal

Navy Children School, Port Blair

Apr 2019 – Mar 2020

Central Board of Secondary Education **Marks: 94%**

Port Blair, Andaman and Nicobar Islands

Extracurricular

Entrepreneurship Development Cell

Feb 2021 – Apr 2024

Executive Head, E-Summit

NIT Durgapur

- Conducted and hosted the second **largest E-Summit** in Eastern India, with more than **15+** events and more than **5,000+** overall participants.